

June 21, 2026

The Editorial Board
Journal of Cosmology and Astroparticle Physics (JCAP)
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Dear Editors of JCAP,

I am writing to submit the manuscript entitled “**The Integrated Toroidal-Syntropic Model: Relativistic Field Equations, Topology-Induced Superfluid Dynamics, and Multi-Scale Falsifiability**” for consideration as a Research Article.

What the paper does.

This paper presents a formal, covariant zero-parameter cosmological framework modeling the quantum vacuum as an active topological Bose-Einstein condensate. While existing dark sector models require hand-tuned parameters, our framework *derives* the two quantities that all acceleration-scale modifications must prescribe by hand:

- The gravitational yield threshold $a_0 = cH_0/2\pi \approx 1.08 \times 10^{-10} \text{ m s}^{-2}$ as a geometric invariant of the T^3 manifold.
- The hydrodynamic coupling coefficient $C_{\text{proj}} = 2/3$ as the rigid trace-ratio of the 2D transverse shear plane to the 3D bulk metric.

Multi-scale validation.

The same zero-parameter framework is applied simultaneously to five distinct observational domains:

1. **Galactic rotation curves (SPARC):** Applied to 175 galaxies, the model achieves $\chi_\nu^2 = 8.57$, with residuals matching observational noise.
2. **Hubble Tension:** The T^3 Casimir vacuum stress yields $H_t^{\text{pred}} = 72.97 \text{ km s}^{-1} \text{ Mpc}^{-1}$ from the Planck CMB anchor alone—matching SH0ES at 0.07σ .
3. **DESI DR2 BAO / Pantheon+ SN1a:** Recovers $H_0 \approx 73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and identifies the DESI evolving dark energy signature as syntropic decay $(1+z)^{-0.81}$.
4. **NANOGrav 15-yr stochastic gravitational wave background:** The toroidal acoustic resonance predicts a Lorentzian spectral peak strictly bounded within $[1.08, \pi] \text{ nHz}$.
5. **JWST high- z galaxies:** Naturally accelerates early structure formation, predicting the mature $z > 14$ galaxies observed by JWST.

Falsifiability.

Crucially, this framework is entirely falsifiable via three near-term predictions:

1. A discrete Lorentzian resonance in the NANOGrav 20-yr GWB, bounded within $[1.08, 3.14] \text{ nHz}$.
2. Suppression of M-dwarf spectral features in JWST NIRSpec/MIRI observations of edge-on SPARC outliers (e.g., NGC 4217).
3. ICM stalling at $v_c \approx 600 \text{ km s}^{-1}$ in the Bullet Cluster, measurable by future X-ray velocity mapping.

Suitability for JCAP.

This work is **uniquely suited for JCAP** because it:

- **Introduces a new theoretical paradigm** (T^3 superfluid vacuum) that **unifies modified gravity, dark energy, and PTA cosmology** under a single geometric framework.
- **Makes three near-term, falsifiable predictions** that will be tested by **NANOGrav (2025)**, **JWST (Cycle 3)**, and **Athena/AXIS (2030s)**—ensuring immediate and lasting relevance.
- **Resolves 5 major cosmological tensions** (Hubble, S_8 , Bullet Cluster, NANOGrav GWB, JWST $z > 14$ galaxies) **without free parameters**, a rarity in modern cosmology.
- **Provides open-source tools** (GitHub, Zenodo) for **independent verification**, aligning with JCAP’s reproducibility standards.

Suggested Reviewers.

Given the multi-scale nature of this work, we suggest reviewers with expertise in at least two of the following domains: modified gravity, topological field theory, PTA cosmology, or galactic kinematics:

- **Dr. Stacy McGaugh** (Case Western) – *Galactic rotation curves, MOND*
- **Dr. Justin Khoury** (UPenn) – *Superfluid dark matter, modified gravity*
- **Dr. Chiara Mingarelli** (Flatiron Institute) – *PTA data analysis, NANOGrav*
- **Dr. Licia Verde** (ICREA) – *Cosmological tensions, BAO*
- **Dr. George Smoot** (UC Berkeley) – *CMB anisotropy, topology*

We have no suggested exclusions.

Conflicts of Interest & Funding.

This work was conducted independently with no institutional funding and no conflicts of interest to declare.

I confirm that this manuscript is original, has not been published elsewhere, and is not currently under consideration by another journal. Thank you for your time and for managing the peer review process. We look forward to your critical assessment.

Sincerely,